A S T A N A I T U N I V E R S I T Y
School of Digital Public Administration

Assignment 1 - Appendix

Business Report — comparative analysis of Company A and Company B

Business Administration (5 ECTS) · Year 1, Trimester 1
IT Management · AI and Business · Digital Public Administration
1st attestation · Due Week 3

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Company A: Ferrero **Company B:** Bayan Sulu
Submission Date: Week 3

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Appendix A — Business Environment Map

Week 1.

	Business Environment
Global	Kazakhstan-based manufacturer; exports to 14 countries
Legal/Political	Joint Stock Company (JSC)
Customers	Consumers in Kazakhstan and nearby countries (children, families, sweet lovers in general)
Technology & AI	Uses automated production equipment and quality-control systems
Competitors	Local chocolate companies, primarily Rakhat
Economic	Sells products through networks, supermarkets, and e-commerce platforms

Appendix B — AI Opportunity Map

Business process	Current issue	Possible AI use	Business value
Commercial sales orders	Salers and distributors submit their orders manually via emails, spreadsheets which leads to delays	AI document processing can be an application that collect and create orders automatically	Faster order processing, more sales
Supplier invoice & price verification	Staff manually check price of chocolate ingredients (sugar, butter, cacao) which is time consuming	AI tool that will automatically check supplier invoices, prices and alerts if prices doesn't match	Cuts down overpricing on raw materials
Selling abroad	Language confusion (Kazakh, Russian and English) for overseas buyers while finding the right recipes or packaging details	An intelligent search tool for foreign buyers. It will allow typing in any language and find flavor, ingredient or exact box size	Faster product discovery for international buyers

Appendix C — Globalization Evidence Task

Globalization question	Your finding about Company B	Evidence / Source
1. Foreign inputs	Does Company B use foreign technology, software, equipment, services or suppliers? **What exactly?** Yes, Bayan Sulu uses foreign technology. In 2011 the factory started up a continuous caramel line from Germany, the STRADA 1200, built by Klöckner Hänsel Processing GmbH. It can turn out 1,200 kg of caramel an hour.	Bayan Sulu company history page [12]
2. International customers/markets	Does Company B sell products/services abroad or serve international customers? **Where?** Yes. Its products go to 14 countries, including China, Mongolia and Korea, and exports make up about 40% of total production.	Bayan Sulu "About us" page (bayansulu.kz) [2]
3. Global competition	Does Company B compete with foreign companies, platforms or products? **Who?** Yes, and most of the pressure comes from abroad. Roughly half of the confectionery on Kazakhstan's market is brought in from other countries, chiefly Russia, Ukraine and Turkey. The company's president has pointed out that imports are the main threat every local producer has to deal with.	The Astana Times, Feb 2018, "Venerable Kazakh confectionery aims to edge out imports with local sweets" [1].
4. Global dependency	Bayan Sulu relies on foreign suppliers in two areas. The first is cocoa. The company buys its flour, sugar and packaging inside Kazakhstan, but the cocoa beans have to be imported, so the prices of its products depend on what those imported ingredients cost. The second is equipment. Its production lines come from foreign manufacturers: Ubitek and Klöckner Hänsel in Germany, FHW in Austria for the waffle line, and Arkan in Turkey for the wrapping machines. The company says it keeps upgrading with machinery from leading European makers.	The Astana Times, Feb 2018 [1] Bayan Sulu, History page [12]
5. Opportunity or Risk	What is the **biggest global opportunity OR risk** for Company B? Why? The main risk is imported cocoa. Chocolate is a core product, so higher world cocoa prices or a weaker tenge hit costs straight away. The president refuses to cut corners on ingredients, unlike competitors who, he says, use flavourings instead of milk and vegetable fats instead of chocolate, and that makes the margin squeeze worse. The main opportunity is halal exports. The company is targeting Central Asia, with high halal demand and over 75 million people, and is also studying the Middle East, China and Southeast Asia. Exports are already about 40% of production, shipped to 14 countries.	The Astana Times article [1]

Appendix D — Stakeholder + AI-Ethics Map

Stakeholder Who is affected	**What they want / risk** What do they want, and what could they risk?	**AI or business issue** What AI or business issue affects them?	**Company B obligation** Company B do to protect or support them?	**Company A evidence** How does Company A handle a similar issue? What evidence did you find?	**Transfer verdict** Should Company B use Company A's practice? Why?
Example:Customers	Fast, fair, safe service	AI customer decisions	Privacy + human review	How company A handles it	ADAPT / PILOT
Customers	Fresh, affordable sweets that are actually in stock — if the AI gets the forecast wrong, shelves either sit with stale candy or run empty, leads to shortage	AI demand forecasting sets store-level supply and shelf freshness	Set expiry/freshness thresholds the AI must respect; audit forecast accuracy regularly	Ferrero's Alloy.ai pilot targeted reducing overstocks [11]	PILOT
Employees	Want job security and clear tasks; risk being blamed for AI mistakes or losing their job	AI decides packing/shipping order based on forecasts, not staff judgment	Require human sign-off before AI decisions take effect; retrain staff to work with the system	Ferrero uses Alloy.ai to give staff full visibility across the supply chain — the AI just provides insight, humans still make the decisions [11].	ADAPT
Suppliers	Predictable order volumes to plan their own production; risk being caught off guard by sudden order changes from the AI.	The AI's demand forecast decides how much raw material gets ordered	Share forecast ranges with key suppliers; give notice before large swings	Ferrero's AI system focuses on understanding demand and working more closely with supply partners [11].	ADAPT

Appendix E — Weekly Record 1

Comparison of Bayan Sulu to Ferrero

Global:

Bayan Sulu is a Kazakhstan company which has international and export activity [1]. It exports to 14 countries [2]. Ferrero is a worldwide company. It operates in 50+ countries and its products are sold in more than 170 countries [3]. This gap in reach shows up in revenue too: for the fiscal year ended August 31, 2025, Ferrero Group reported consolidated turnover of €19.3 billion (up 4.6% year-on-year) and more than 50,000 employees worldwide [3], while JSC Bayan Sulu reported 2025 revenue of 51.79 billion KZT (≈$114 million, up 9.53% from 47.28 billion KZT in 2024) [4].

Legal policy: Bayan Sulu's legal form is JSC- Joint Stock Company [2]. Ferrero's legal form is Ferrero International S.A, which is a Public Limited Company under Luxembourg law [5]. Since Ferrero operates globally, individual Ferrero subsidiaries can have different legal forms.

Customers: Bayan Sulu and Ferrero have similar customers because both sell confectionery products. Bayan Sulu mainly serves customers in Kazakhstan and nearby countries**,** including children, families, and people who enjoy sweets such as chocolate, biscuits, and candies**.** Ferrero has a much wider global customer base. Its products are popular among children, families, and adults in many countries. Ferrero focuses on well-known brands such as Nutella, Kinder, and Ferrero Rocher.

Technology/AI: Bayan Sulu uses modern production technologies to make confectionery products, including automated equipment and quality-control systems. It also uses digital technologies to improve production and business processes. Ferrero uses more advanced technologies**,** including automation, data analysis, digital systems, and AI in some business and production processes. AI can help Ferrero improve production, quality control, supply chains, and understand customer needs.

Economy: Bayan Sulu mainly sells its products through supermarkets, shops, distributors, and online platforms in Kazakhstan and other countries. Its products are affordable and aimed at a wide range of customers. Ferrero sells its products through supermarkets, retail stores, distributors, and online shopping platforms around the world. Its well-known brands are available in many countries, which helps the company reach a large global market. Both companies grew in their most recently reported year: Bayan Sulu's net income rose 28.41% to 4.44 billion KZT in 2025 [4], reflecting healthy profitability from its affordable, mass-market pricing, while Ferrero's 4.6% turnover growth points to continued demand for its premium, brand-led products [3].

Competitors: Bayan Sulu's main competitors are LOTTE Rakhat (formerly Rakhat, majority-owned by South Korea's Lotte Confectionery since 2013) [6] and other local confectionery companies in Kazakhstan. Ferrero competes with large international companies such as Mondelez, Mars, Nestlé, and Lindt**.** Ferrero has stronger competition in the global market because its products are sold in many countries. The scale of that competition is significant: Mondelez reported net revenue of $36.44 billion for fiscal year 2024 [7], Nestlé reported total sales of CHF 91.4 billion for 2024 [8], and Lindt & Sprüngli reported total sales of CHF 5.92 billion for fiscal year 2025 [9]. Mars is privately held and does not publish detailed public financial reports, but it is widely regarded as being of a similar scale [10].

How we can enable AI

AI can help to solve possible problem of understanding customer demand and managing production efficiently. AI could analyze previous sales, seasonal demand, and customer preferences to predict which products will be more popular. This could help Bayan Sulu produce the right amount and reduce excess products.

Recommendations

Expand export markets gradually: Bayan Sulu currently exports to only 14 countries and around 40% of its output [2], compared with Ferrero's presence in over 170 markets [3]. Growing further in nearby CIS, Central Asian, and Gulf markets, where Bayan Sulu already has export activity, would be a lower-risk way to increase that share before pursuing more distant markets.

Pilot AI-driven demand forecasting: Ferrero USA has adopted an AI-based supply-chain platform (Alloy.ai) to improve demand forecasting and reduce overstock and stockouts [11]. Bayan Sulu could pilot a similar, smaller-scale forecasting tool to support the demand-prediction use case described above and reduce excess production.

Build a focused international brand portfolio: Ferrero's global reach rests on a small number of flagship brands (Nutella, Kinder, Ferrero Rocher) rather than its full product range. Bayan Sulu could concentrate export marketing on two or three flagship product lines to build recognition beyond Kazakhstan.

Pursue international quality certifications: Competing for shelf space against Mondelez, Nestlé, Lindt, and Ferrero in export markets will likely require additional international food-safety and quality certifications (e.g., ISO 22000, HACCP) and export-ready packaging and labeling.

Consider strategic partnerships: Bayan Sulu's closest local competitor, LOTTE Rakhat, gained access to regional distribution and capital after Lotte Confectionery (South Korea) acquired a majority stake in 2013 [6]. A similar strategic partnership or joint venture could give Bayan Sulu comparable resources for international expansion.

Appendix F — Weekly Record 2

Artifact: Stakeholder and AI-Ethics Map — Bayan Sulu and Ferrero

This artifact uses two simple ideas together. A *stakeholder map* shows the different groups a company affects — customers, employees, suppliers, investors, regulators, and the environment. An *AI-ethics lens* then looks at each group and asks: what could go wrong, or what should the company be careful about, if it uses AI here? Bayan Sulu is a Kazakhstan company that sells to 14 countries and mostly uses automation to help make and check its products [2]. Ferrero is a much bigger, family-owned company that sells in over 170 countries and uses AI in more parts of its business, such as predicting demand and checking its supply chain [3].

Customers / Consumers — *main issue: privacy and honesty.* Bayan Sulu does not use much AI directly with customers; most of its automation is in the factory, for making and checking products [2]. If it starts using AI to predict demand, it should make sure this does not change prices or labels in a way that misleads customers. Ferrero interacts with customers more directly, through online shopping, loyalty programs, and AI tools that predict demand, such as the Alloy.ai system used by its U.S. team [11]. Because Ferrero collects more customer data, it has a bigger responsibility to protect that data.

Employees — *main issue: fairness and human oversight.* Bayan Sulu's automated machines affect jobs at one factory, so the company should help retrain or move workers as more tasks become automated. Ferrero uses automation and AI in many more factories and supply-chain jobs around the world [3]. Because the effect is bigger, Ferrero needs stronger, more consistent training programs, and must make sure that if it ever uses AI to help with hiring or reviewing employees, the AI treats workers fairly no matter which country they are in.

Suppliers / Producers — *main issue: accountability.* Bayan Sulu mainly buys from local and regional suppliers. If it starts using AI to predict how much it needs to order, it should avoid sudden changes that could hurt small suppliers. Ferrero buys ingredients like palm oil, cocoa, and hazelnuts from all over the world. It uses a satellite-based AI tool called "Starling," built with the Earthworm Foundation and Airbus, to check whether its palm oil suppliers are cutting down forests [13]. This is a good example of AI helping with ethics — but only if Ferrero actually acts on what the AI finds, not just collects the data.

Investors / Shareholders — *main issue: openness.* Bayan Sulu is listed on the Kazakhstan Stock Exchange, so it must follow normal rules about telling investors what it is doing, including any AI projects [2]. Ferrero is privately owned by the Ferrero family, not publicly traded [3]. This means Ferrero can choose how much it tells the public about its AI use and risks. Bayan Sulu's investors get more automatic information; Ferrero's openness depends on the company's own choice.

Regulators / Government — *main issue: following the rules in each country.* Bayan Sulu only has to follow Kazakhstan's food-safety and stock-market rules. Ferrero sells in more than 170 countries, so it must follow many different rules at once [3], including new AI rules like the European Union's AI Act. An AI tool that is fine in one country might not meet the rules in another, so Ferrero has a much bigger job keeping track of this.

Environment / Local Communities — *main issue: using AI to help the environment.* Bayan Sulu's environmental impact is local, tied to its one factory in Kazakhstan. Ferrero's impact is global, and it uses AI and satellite monitoring to watch for deforestation in its palm-oil supply chain [13]. This shows that AI can do more than create risk — it can also be a tool that helps protect the environment. But

bigger companies like Ferrero also carry more responsibility, because they can cause more harm if they do not act on what the AI shows them, and they also have more resources to invest in this kind of monitoring

Recommendations for Bayan Sulu on AI Ethics

Based on the stakeholder map above, here are practical steps Bayan Sulu could take now, while its use of AI is still small, so that ethical safeguards grow alongside the technology rather than being added after a problem happens.

Protect small suppliers from sudden changes: If Bayan Sulu uses AI to forecast how much it needs to order, it should give local suppliers advance notice of demand changes rather than sudden ones, so that small regional suppliers are not hurt by AI-driven decisions they cannot predict.

Plan for the effect on workers early: As more factory tasks become automated, Bayan Sulu should offer training programs so employees can move into new roles instead of losing their jobs. Planning this before automation expands is easier than reacting to job losses afterward.

Set clear rules for data before AI use grows: Bayan Sulu should decide now how any customer or business data used by AI will be stored and protected, even while its AI use is still limited. This is much easier to do early than to fix later, once far more data is involved.

Borrow Ferrero's supply-chain idea, at a smaller scale: Ferrero uses satellite-based AI monitoring to check whether its palm-oil suppliers are causing deforestation [13]. Bayan Sulu could use a simpler version of this idea — basic data tools to check that its own suppliers meet quality and fair-labor standards — instead of relying only on trust.

Write a short, plain AI-ethics policy: A written policy would help — for example, stating that no AI decision is made without a person checking it, and that customer data is never used by AI without customers being told. A simple written policy like this would put Bayan Sulu ahead of many similarly sized companies and would make future growth in AI use much easier.

Appendix G — Evidence Folder

(IEEE numbered list, current to Week 2)

This folder contains all credible sources collected and verified across

Weeks 1–2. Sources are numbered in order of first appearance in the Weekly

Records and the Business Report.

References

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